PES UNIVERSITY

Department of Computer Science and Engineering

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #47
Domain & SSL Certificate Expiry Alert System

Name	BHARATH K
SRN	PES2UG24CS113
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Final Submission Document

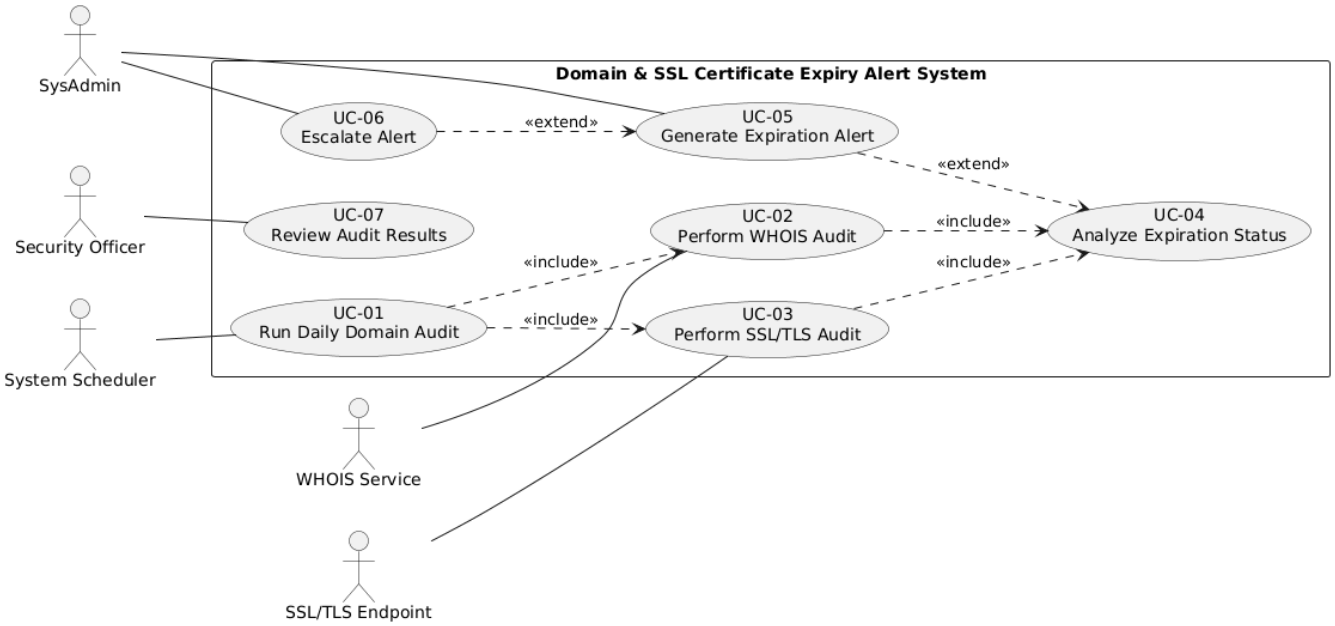
1. Requirements Table

The following requirements define the functional and nonfunctional requirements for the Domain & SSL Certificate Expiry Alert System.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall initiate TLS handshakes against monitored domains daily, extract certificate expiration dates, and trigger alerts at 30, 15, and 3 days before expiry.	High	Pass: An alert is queued when a monitored certificate reaches 30, 15, or 3 days before expiry. Fail: A certificate reaches any threshold without an alert.	Prevents unexpected SSL/TLS certificate expiration and service disruption.
FR-002	Functional	The system shall perform a daily WHOIS audit for all monitored domains and retrieve their domain registration expiration dates.	High	Pass: The system retrieves and records the registration expiration date for every valid monitored domain. Fail: Any monitored domain is not audited or its expiration date is not recorded.	Enables early detection of domains approaching registration expiration.
FR-003	Functional	The system shall compare the retrieved domain and certificate expiration dates against their configured alert thresholds and identify assets requiring attention.	High	Pass: Every monitored domain or certificate reaching its configured threshold is identified. Fail: An asset meeting a threshold is not identified.	Ensures expiration risks are detected consistently before they become critical.
FR-004	Functional	The system shall send expiration alerts to the SysAdmin according to the configured notification and escalation procedure.	High	Pass: When an expiration threshold is reached, the alert is delivered to the designated SysAdmin. Fail: The alert is generated but not delivered.	Allows corrective action before domain or certificate expiration.
FR-005	Functional	The system shall allow the Security Officer to review the latest WHOIS and SSL/TLS audit results and the expiration status of monitored domains and certificates.	Medium	Pass: The Security Officer can retrieve and view the latest audit result and expiration status for each monitored asset. Fail: The information cannot be viewed.	Provides visibility into the security and operational status of monitored assets.
NFR-001	Nonfunctional	The monitoring engine shall scan a list of 1,000 domains and SSL endpoints in under 3 minutes.	High	Pass: Benchmarking confirms that 1,000 domains and SSL endpoints are scanned in less than 3 minutes under simulated peak load. Fail: The scan takes 3 minutes or longer.	Ensures the daily monitoring process completes efficiently at the specified scale.
NFR-002	Nonfunctional	The system shall retain the results of each completed daily WHOIS and SSL/TLS audit so that the latest audit status can be reviewed without performing the audit again.	Medium	Pass: Completed audit results can be retrieved without initiating a new audit. Fail: Results cannot be retrieved without repeating the audit.	Provides reliable access to audit results for operational and security review.

2. UML Use-Case Diagram

Figure 1. UML Use-Case Diagram — Domain & SSL Certificate Expiry Alert System



3. Use-Case Flow Specification

Use Case ID	UC-01
Use Case Name	Run Daily Domain Audit
Primary Actor	System Scheduler
Supporting Actors	WHOIS Service, SSL/TLS Endpoint, SysAdmin
Goal	Perform the daily WHOIS and SSL/TLS audits and identify domains or certificates approaching expiration.

3.1 Preconditions

• The monitoring system is operational.
• The domains and SSL/TLS endpoints to be monitored are configured in the system.
• The System Scheduler initiates the scheduled daily audit.

3.2 Postconditions

Success: The latest WHOIS and SSL/TLS audit results and expiration status are recorded, and required alerts are generated.
Failure: The affected audit is recorded as unsuccessful and the remaining monitored assets can continue to be processed.

3.3 Main Success Scenario

1. The System Scheduler initiates the daily domain audit.
2. The system retrieves the list of monitored domains and SSL/TLS endpoints.
3. The system performs a WHOIS audit for each monitored domain.
4. The WHOIS Service returns the domain registration information, including its expiration date.
5. The system records the domain registration expiration date.
6. The system initiates an SSL/TLS handshake with each monitored SSL/TLS endpoint.
7. The system extracts the certificate expiration date from the TLS certificate.
8. The system analyzes the domain and certificate expiration dates against the configured alert thresholds.
9. If an expiration threshold is reached, the system generates an expiration alert.
10. The system applies the configured escalation procedure and sends the required alert to the SysAdmin.
11. The system records the completed audit results for later review.
12. The daily audit is completed successfully.

3.4 Alternate Flow — Certificate Approaching Expiration

9a1. During expiration analysis, the system detects that a monitored certificate is approaching expiry.
9a2. If the certificate reaches 30, 15, or 3 days before expiry, the system triggers the corresponding expiration alert.
9a3. The alert is sent to the SysAdmin according to the configured notification and escalation procedure.
9a4. The system records the alert together with the audit result.
9a5. The system continues processing the remaining monitored domains and SSL/TLS endpoints.